

SUPPLEMENTARY MATERIAL TO COGNITIVE WARRANT RESEARCH

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This document supplements my research documents, providing background information, intermediate steps, or detailed explanations for those that are interested. It is a work in progress.

The concept of information is too broad to be captured completely by a single definition. However, for any probability distribution, we define a quantity called the *entropy*, which has many properties that agree with the intuitive notion of what a measure of information should be. This notion is extended to define *mutual information*, which is a measure of the amount of information one random variable contains about another. Entropy then becomes the self-information of a random variable. Mutual information is a special case of a more general quantity called *relative entropy*, which is a measure of the distance between two probability distributions. All these quantities are closely related and share a number of simple properties....

...these quantities arise as natural answers to a number of questions in communication, statistics, complexity, and gambling. That will be the ultimate test of the value of these definitions.

—Thomas M. Cover & Joy A. Thomas,
Elements of Information Theory (2006, p. 13)

CHAPTER 1: Probability and Information Theory

1.1. Conditional Probability

Fundamental terms of probability

- An **experiment** is any action or process whose outcome is subject to uncertainty.
- The **sample space** is the set of all possible outcomes of that experiment.
- An **event** is any collection (subset) of outcomes contained in the sample space.
- An event is **simple** if it consists of exactly one outcome and **compound** if it consists of more than one outcome.
- A and B are considered **independent** events if the occurrence or nonoccurrence of one event has no bearing on the chance that the other will occur.

We intuitively know that a probability is some measure of an event's likelihood, or the certainty of an outcome, or our degree of belief in a proposition or hypothesis. The probability of an event or outcome is 0, 1, or any number between.

If we have defined an event A , then $P(A)$ is the marginal, or unconditional, probability of event A occurring without consideration of any other events. $P(B)$ is thus the marginal probability of event B . If A and B are independent, as defined above, then knowing whether or not B has occurred, or been observed, provides no information about, or has no effect on, the subsequent (non)occurrence of A . If, however, these two events are *not* independent, then the occurrence of B does provide

information about, or has an effect on, the subsequent occurrence or observation of A . In this case, the probability of A is conditioned by B and is notated as $P(A | B)$.

Definition 1.1 (Conditional probability). Let A and B be two events with $P(B) \neq 0$. The conditional probability of A given B is defined to be

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad (1.1)$$

The notation $P(A \cap B)$ represents the probability of *both* events A and B occurring.

As an example, the (unconditional) probability that a person is wet at any moment might be $P(A) \geq 0$ and the (unconditional) probability that it is raining at that person’s location at any moment might be $P(B) \geq 0$. If we know that B has occurred or is occurring, the probability that this person is wet changes. In this situation, the person’s state of being drenched is conditional upon B having occurred. We will see below that, on average, conditioning (i.e., knowing the state of one random variable) reduces some uncertainty about another random variable (but leaves that uncertainty unchanged if those random variables are independent). If I know the result of flipping one coin, that does not affect my uncertainty about the result of flipping a different coin; however, if I know that it is raining at a person’s location, that reduces my uncertainty (to some degree) about whether he or she is wet.

1.2. Entropy

A **random variable** is a quantity whose value depends on the outcome of an uncertain process. The name is slightly misleading — it doesn’t mean arbitrary or haphazard. It means the value is determined by something not yet observed.¹ A coin

¹Indeed, Deisenroth et al. (2020) note that “the name ‘random variable’ is a great source of misunderstanding as it is neither random nor is it a variable. It is a function.”

flip, a person's response time on a trial, and whether two sentences invoke the same word sense are all random variables: before you observe them, they could take more than one value; after you observe them, they take exactly one. The possible values of a **discrete random variable** are finite or "countably infinite." An example of a discrete random variable is the sum of rolling two dice, as its possible values are the integers 2 through 12. What makes a random variable mathematically tractable is that even though individual outcomes are uncertain, the *distribution* of outcomes — which values are likely and which are unlikely — can be known and reasoned about precisely.

Entropy measures how uncertain that distribution is. More precisely, it quantifies how surprised you should expect to be by the outcome — or equivalently, how much information you gain when you learn it. A fair coin has high entropy: you genuinely don't know what's coming, so the outcome tells you something. A coin weighted to land heads 99% of the time has low entropy: you already know roughly what will happen, so the outcome tells you little. Crucially, entropy is a property of the *distribution*, not of any single observation and not of what the outcomes actually are.

You may be familiar with the term entropy from the second law of thermodynamics and some version of 'over time, everything becomes more disordered.' I won't discuss the use of this term in thermodynamics versus information theory, but perhaps associating disorder with uncertainty will make it easier to understand what is meant by entropy here.

Definition 1.2 (Entropy). Formally, the entropy of the random variable X is

$$H(X) = - \sum_x p(x) \log p(x). \quad (1.2)$$

For each of the possible values of the random variable, we are multiplying its probability $p(x)$ by the logarithm² of its probability $\log p(x)$. The entropy $H(X)$ is the negated sum of these products.

Earlier, I said that entropy quantifies how surprised you should expect to be by the outcome of an uncertain process. Surprise is the result of an unexpected event, or, in other words, an event with a low probability. Thus, *surprise and likelihood are inversely related*: the greater the likelihood of an event, the less surprise when it occurs.

A property of logarithms is $\log M^k = k \cdot \log M$. Because of the distributive property, we can bring the negative sign into our summation, and Equation (1.2) can be written as

$$\begin{aligned} H(X) &= \sum_x -p(x) \log p(x) \\ &= \sum_x p(x) \cdot (-1) \cdot \log p(x) \\ &= \sum_x p(x) \cdot \log(p(x))^{-1} \\ &= \sum_x p(x) \cdot \log \frac{1}{p(x)} \end{aligned}$$

²Note that the base of the logarithm is arbitrary. Base 2 is standard in information theory and the units of entropy are bits for “binary digits.”

The last term $\frac{1}{p(x)}$ is the inverse of the likelihood of x and thus encodes surprisal in entropy. Taking the logarithm transforms the scale but not the inverse relationship. We can now say that for each possible value of X , we are multiplying the surprise associated with that value by the value's likelihood. Entropy is the sum of these products. You may be familiar with a weighted sum. The expected value of a random variable is the sum of its possible values each weighted by their probability. Entropy is thus the expected amount (or weighted average) of surprise over the distribution of the random variable.

An immediate consequence of Definition 1.2 is $H(X) \geq 0$: entropy is always non-negative. If the outcome is certain, $H(X) = 0$; there is nothing to learn. Otherwise $H(X) > 0$; some uncertainty always remains.

Figure 1.1 shows entropy for the simplest possible case: a Bernoulli variable that takes only two values (such as 0 or 1), with probability p for one outcome and $(1 - p)$ for the other. Entropy is zero at the extremes — when the outcome is a near-certainty in either direction ($p = 0$ or $p = 1$) — and peaks at $p = \frac{1}{2}$, when both outcomes are equally likely and uncertainty is at its maximum. The shape of this figure is important in the “H3 as a Mathematical Consequence of the CI Framework” section of my research document.

1.3. Conditional Entropy

Conditional entropy measures how much uncertainty remains about one random variable once the value of another is known, averaged over the values that other variable can take.³ If knowing X tells you a lot about Y , the conditional entropy

³This and following sections are based primarily on *Elements of Information Theory* (2nd ed.) by Cover and Thomas.

Binary Entropy Function

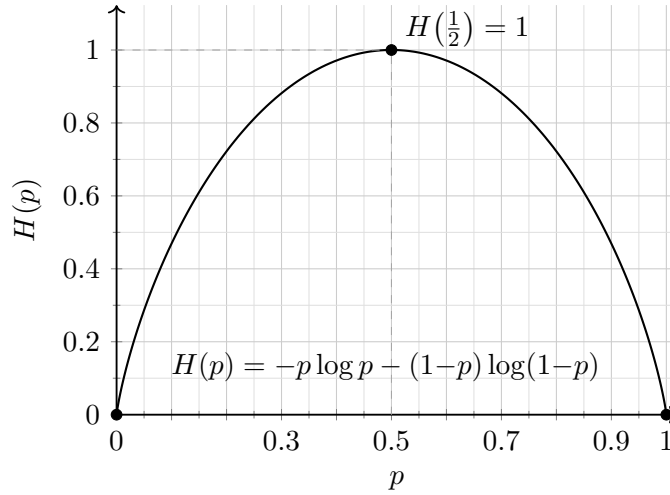


Figure 1.1: Entropy of a two-outcome variable as a function of the probability p of one outcome. Uncertainty — and therefore entropy — is zero when the result is a near-certainty and is maximized when both outcomes are equally likely.

$H(Y | X)$ is low — little uncertainty is left. If knowing X tells you nothing about Y , $H(Y | X)$ equals the original entropy of Y — knowledge of X bought you nothing.

The definition computes this by asking: for each possible value of X , how uncertain is Y ? It then averages those uncertainties, weighted by how often each value of X actually occurs:

$$H(Y | X) = \sum_x p(x) H(Y | X=x) \quad (1.3)$$

$$= - \sum_x p(x) \sum_y p(y | x) \log p(y | x) \quad (1.4)$$

$$= - \sum_x \sum_y p(x, y) \log p(y | x). \quad (1.5)$$

where $p(x, y)$ is the joint distribution.

In the Theoretical Framework chapter of my research document, I introduce the binary random variable Y indicating whether a word used in two different contexts instantiates the same sense of the word, and the continuous quantity semantic distance, denoted d , representing the “dissimilarity” of the two contexts. The conditional probability $P(Y \mid d)$ thus defines a mapping from semantic distance to sense identity. At low distances the mapping is highly predictable toward same-sense identity; at high distances, it is equally predictable toward different-sense identity. At intermediate distances the mapping becomes uncertain, which can be formalized using **conditional entropy**:

$$H(Y \mid D=d) = - \sum_{y \in \{0,1\}} p(y \mid d) \log p(y \mid d) \quad (1.6)$$

You may have noted an apparent discrepancy between Equation (1.6) and Equation (1.4): there is no weighting by how often each value of d occurs. The resolution is that the two equations denote different objects. Equation (1.4) is the **averaged** conditional entropy $H(Y \mid D)$: a single number, obtained by averaging the residual uncertainty in Y across the entire distribution of distances. Equation (1.6) is the **pointwise** conditional entropy at one particular distance, written here as $H(Y \mid D=d)$. It is not a number but a function of d — the entropy of a single conditional distribution. No weighting appears because no averaging has been performed. The two are related by an expectation over the distribution of distances, which for continuous d with density $f(d)$ is

$$H(Y \mid D) = \mathbb{E}_D[H(Y \mid D=d)] = \int f(d) H(Y \mid D=d) dd. \quad (1.7)$$

It is tempting to attribute the difference instead to the fact that semantic distance functions as a controlled independent variable in my experimental design rather

than as a “random” variable in the sense discussed above. That framing does not account for the missing weights. A controlled variable still has an empirical design distribution over which one could average, and conversely one can always condition on a single realized value of a genuinely random variable. The operative distinction is between conditioning on a particular value and averaging across values — not between fixed and random quantities.

The pointwise form is the one the framework requires. Cognitive information is defined at each distance, $CI(d) \equiv I(Y; S \mid d)$, and Hypothesis 3 concerns the *shape* of that function across the range of d — a claim that presupposes a quantity varying with distance rather than a scalar summary. The aggregate is recovered when needed by the same expectation used in Equation (1.7). Throughout, I write $H(Y \mid D=d)$ when conditioning on a particular value and $H(Y \mid D)$ for the average.

$H(Y \mid D=d)$ represents the disambiguation load. My claim is that cognitive processing difficulty in the primed sensibility judgment task of Experiment 1 increases with this uncertainty: higher $H(Y \mid D=d)$ corresponds to longer response times as the cognitive system labors to resolve which sense is intended.

Note. The order of conditional entropy matters: knowing X does not generally reduce uncertainty about Y by the same amount that knowing Y reduces uncertainty about X , so $H(Y \mid X) \neq H(X \mid Y)$ in general. However, the *difference* each variable makes is symmetric: $H(X) - H(X \mid Y) = H(Y) - H(Y \mid X)$. Both expressions equal the mutual information between X and Y — the uncertainty each one resolves about the other — introduced in Section 1.5.

1.4. Relative Entropy

Information theory provides a measure of the distance between two distributions.

Definition 1.3. The relative entropy or Kullback-Leibler (KL) divergence between two probability mass functions $p(x)$ and $q(x)$ is defined as

$$D(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \quad (1.8)$$

We will use the conventions that $0 \log \frac{0}{0} = 0$, $0 \log \frac{0}{q} = 0$, and $p \log \frac{p}{0} = \infty$.

There are several interpretations of this concept that can be useful later. The original motivation was as an expected log likelihood ratio as a way to distinguish between two distributions, i.e. a measure of how much an approximating probability distribution Q is different from the true distribution P .

Another interpretation is as the information gain achieved if the true distribution P would be used instead of the currently used Q ; conversely, it can be interpreted as the amount of information lost when Q is used to approximate P . Put in terms of Bayesian inference, relative entropy is a measure of the information gained by revising one's beliefs from the prior probability distribution Q to the posterior probability distribution P . We will be using this last interpretation in our data analysis.

Note that relative entropy is not a true distance between distributions since it is not symmetric ($D(p \parallel q) \neq D(q \parallel p)$ in general) and does not satisfy the triangle inequality (including a third distribution u , $D(p \parallel q) \leq D(p \parallel u) + D(u \parallel q)$ does not hold in general). Although relative entropy is technically a “divergence” and not a distance, it is often useful to think of it as a “distance” between distributions.

1.5. Mutual Information

Mutual information $I(X; Y)$ measures how much knowing one variable tells you about another. Equivalently, it is the reduction in uncertainty about X that

results from learning Y — or vice versa, since the relationship is symmetric. Two variables that are completely unrelated have zero mutual information; the more tightly their values are linked, the higher it is.

$$I(X; Y) = D(p(x, y) \parallel p(x)p(y)) \quad (1.9)$$

$$= \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \quad (1.10)$$

The formula compares the actual joint distribution of X and Y — how they behave together — to what their joint distribution *would* be if they were completely independent. When the two distributions match, i.e. the random variables *are* independent, the variables share no information and $I(X; Y) = 0$. The more the true and independent distributions diverge, the more the variables mutually provide information about each other.

1.6. Relationship Between Entropy and Mutual Information

Entropy, conditional entropy, and mutual information are not separate concepts — they are different ways of slicing the same underlying uncertainty (or potential information). The key identity is:

$$I(X; Y) = H(X) - H(X | Y). \quad (1.11)$$

In words: mutual information is exactly the uncertainty about X that disappears once you know Y . Or, rearranging Equation 1.11,

$$H(X) = I(X; Y) + H(X | Y). \quad (1.12)$$

Everything that was uncertain about X either (a) gets resolved by learning Y , which is the mutual information, or (b) remains uncertain even after learning Y , which is the conditional entropy. Nothing is double-counted; nothing is lost.

Because mutual information is symmetric — X tells you as much about Y as Y tells you about X — the same decomposition runs in both directions:

$$I(X; Y) = H(Y) - H(Y | X). \quad (1.13)$$

A third way to express the same relationship uses joint entropy $H(X, Y)$, the total uncertainty when X and Y are considered together:

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (1.14)$$

If X and Y were independent, their joint uncertainty would equal the sum of their individual uncertainties. Mutual information is the amount by which the joint uncertainty falls *short* of that sum — the overlap that would be double-counted if the marginals were naively added (see Figure 1.2).

Collecting these results:

$$I(X; Y) = H(X) - H(X | Y) \quad (1.15)$$

$$= H(Y) - H(Y | X) \quad (1.16)$$

$$= H(X) + H(Y) - H(X, Y) \quad (1.17)$$

$$= I(Y; X) \quad (1.18)$$

$$I(X; X) = H(X) \quad (1.19)$$

The last line has an intuitive reading: a variable contains exactly one variable’s worth of information about itself — its own full entropy, no more and no less — which is why this document’s epigraph referred to entropy as self-information.

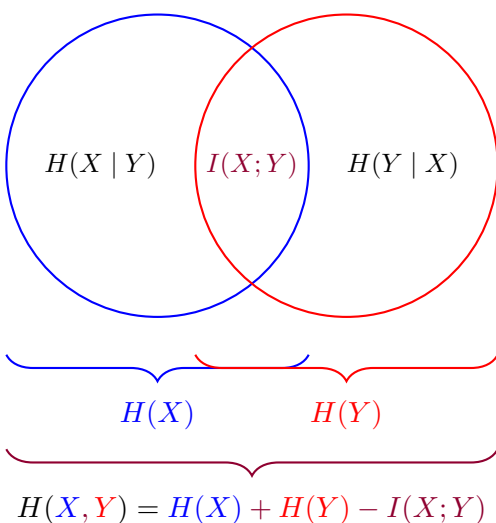


Figure 1.2: Entropy and mutual information as overlapping regions of $H(X)$ and $H(Y)$. $I(X;Y)$ is the shared overlap — uncertainty that either variable resolves about the other. The conditional entropies $H(X|Y)$ and $H(Y|X)$ are the portions exclusive to each variable. The joint entropy $H(X,Y)$ is the total area of both circles combined (*without counting the mutual information twice*).